

Decision-Support Application

ELECTRE II and IV Methods

End-of-Studies Internship Defense

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1. Context & Institutional Mission

- **Host Institution:** University of West Attica (Athens, Greece).
- **Duration:** 12 weeks (April 07, 2026 — June 30, 2026).
- **The Mission:** Designing and implementing a Multicriteria Decision Analysis (MCDA) application completely from scratch.
- **Operational Need:** Streamline complex business choices where competing options oppose each other across heterogeneous and conflicting dimensions.

2. The Theoretical Challenge: MCDA Schools

- **The American School (Compensatory):**
 - Aggregates all criteria into a single utility function.
 - *Risk:* A single excellent score masks an absolute deficiency.
- **The European School (Non-Compensatory):**
 - Based on the concept of **Outranking** (aSb).
 - Natively integrates indifference, incomparability, and strict veto thresholds.
 - **Target Paradigms:** ELECTRE II (Weighted) & ELECTRE IV (Weightless).

3. Algorithmic Processing Core

ELECTRE II: Linear Normalization & Distillations

$$n_j(a) = \frac{g_j(a) - \min(g_j)}{\max(g_j) - \min(g_j)}$$

- Calculates Global Concordance ($C(a, b)$) vs. Discordance ($D(a, b)$).

ELECTRE Iv: Strict Non-Compensatory Vetoes

- Validates outranking without relative criteria coefficients.
- Activation of the absolute veto barrier :

$$q_{j^*}(b) - q_{j^*}(a) > v_{j^*}$$

4. Functional Workflow & Safeguards

- **Matrix Lifecycle Controls:**

- Strict physical lock preventing the matrix from dropping below a 2x2 layout.

- **Real-Time Integrity Engine:**

- Key-by-keystroke interception of empty or non-numeric cells.
- Calculation pipeline suspended instantly upon invalid input detection.

- **Polymorphic Layout Switching:**

- Adaptive DOM restructuring based on the active calculation engine.

5. Software Architecture Patterns

- **Strict Layer Isolation:**

- Core mathematical logic encapsulated into pure function modules.
- Zero knowledge of graphics, rendering, or the DOM state.

- **Technical Stack Choices:**

- **TypeScript & Node.js:** Compile-time static typing of multi-dimensional arrays.
- **Frameworkless Approach:** Lightweight architecture using raw DOM manipulation to eliminate external overhead.

6. Engineering Pitfalls & Optimizations

- **DOM Lifecycle Resolution:**

- Avoided local event binding issues by leveraging native event bubbling via a unified delegation hub.

- **Algorithmic Hoisting ($O(N^3) \rightarrow O(N^2)$):**

- Amplitude calculation moved outside the nested comparison loops.

- **Defensive Arithmetic Gates:**

- Prevention of division-by-zero on homogeneous criteria via automatic fallback mechanisms.

7. Practical Validation: Case Study

- **Scenario:** Procuring a corporate smartphone fleet (3 alternatives, 3 criteria).
- **ELECTRE II Output:** Confirms clear dominance graphs but flags cost-based vulnerabilities.
- **ELECTRE Iv Output:** Dynamic adjustment of inputs, activation of vetoes, and revelation of the strict outranking network.
- **Performance:** Fully stable execution optimized for in-memory processing.

Conclusion & Perspectives

- **Technical Takeaways:** Practical mastery of modern software patterns and advanced static typing paradigms.
- **Human Takeaways:** Deepening international autonomy and technical decision-making within a foreign institution.
- **Future Evolutions:** Dynamic graph visualization modules and automated PDF/LaTeX results exporting.